Literature Review on Mavacamten (Camzyos):

Drug Development, Clinical Trials, Therapeutic Applications,

and Market Impact

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Submitted to:

CodeAlpha – Pharmaceutical Research Internship Program

Date: July 2026

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1 Abstract

Hypertrophic cardiomyopathy (HCM) is one of the most common inherited cardiovascular diseases and is characterized by myocardial hypertrophy, impaired ventricular relaxation, hypercontractility, left ventricular outflow tract (LVOT) obstruction, arrhythmias, and an increased risk of sudden cardiac death. For decades, treatment primarily relied on symptom management using beta-blockers, calcium channel blockers, and invasive septal reduction procedures. The development of mavacamten (Camzyos), the first FDA-approved cardiac myosin inhibitor, represented a major shift toward mechanism-based therapy by directly targeting sarcomeric hypercontractility. This literature review summarizes the drug discovery process, molecular mechanism, pharmacokinetics, preclinical evidence, major clinical trials, therapeutic applications, safety profile, and commercial significance of mavacamten in obstructive hypertrophic cardiomyopathy.

2 Introduction

Hypertrophic cardiomyopathy affects approximately one in every 500 individuals worldwide and is primarily caused by mutations in genes encoding sarcomeric proteins. These mutations increase myocardial contractility, impair ventricular relaxation, and promote dynamic obstruction of the left ventricular outflow tract. Patients commonly experience dyspnea, chest pain, syncope, exercise intolerance, heart failure symptoms, and may be at risk for sudden cardiac death.

Conventional pharmacological therapy has traditionally focused on symptom control through negative inotropic agents such as beta-blockers, verapamil, and disopyramide. Although these medications reduce symptoms, they do not directly modify the molecular mechanisms responsible for disease progression. Consequently, many patients eventually require septal myectomy or alcohol septal ablation.

The discovery of cardiac myosin inhibitors introduced a new therapeutic strategy aimed at correcting the underlying sarcomeric dysfunction. Among these agents, mavacamten became the first disease-specific pharmacological treatment approved for symptomatic obstructive hypertrophic cardiomyopathy.

3 Literature Review

3.1 Pathophysiology of Hypertrophic Cardiomyopathy

Hypertrophic cardiomyopathy is characterized by excessive myocardial contractility resulting from mutations affecting sarcomeric proteins. Rather than simple myocardial thickening, the disease involves abnormal cross-bridge cycling between actin and myosin, increased ATP consumption, impaired ventricular relaxation, and inefficient cardiac energetics. These abnormalities produce dynamic LVOT obstruction and progressive functional impairment.

Earlier treatment strategies primarily targeted symptom relief without correcting these molecular abnormalities. This limitation motivated researchers to identify therapies capable of directly regulating sarcomeric function.

3.2 Drug Discovery and Molecular Mechanism

The development of mavacamten originated from research demonstrating that cardiac myosin could serve as a therapeutic target. Mavacamten is a selective, reversible, allosteric inhibitor of beta-cardiac myosin ATPase.

Instead of blocking calcium channels or adrenergic receptors, mavacamten decreases the number of myosin heads available for actin interaction by stabilizing the super-relaxed state of myosin molecules. As fewer cross-bridges are formed, myocardial contractility decreases, reducing LVOT obstruction and improving ventricular filling.

This mechanism directly addresses the molecular cause of obstructive hypertrophic cardiomyopathy rather than simply reducing symptoms.

3.3 Preclinical Studies

Animal experiments and myocardial tissue studies demonstrated favorable pharmacokinetic and pharmacodynamic properties. Mavacamten exhibited:

- High oral bioavailability
- Long elimination half-life
- Low systemic clearance
- Dose-dependent reduction of myocardial contractility
- Reduced calcium sensitivity
- Improved energetic efficiency

These findings supported progression toward human clinical trials.

3.4 Clinical Development

Clinical development progressed through standard Phase I, II, and III studies.

3.4.1 Phase I

Phase I trials evaluated pharmacokinetics, safety, and tolerability in healthy volunteers. Investigators demonstrated rapid absorption, acceptable safety, and identified CYP2C19 genotype as an important determinant of drug exposure.

3.4.2 Phase II: MAVERICK-HCM

The MAVERICK-HCM trial investigated mavacamten in patients with symptomatic non-obstructive HCM.

Important findings included:

- Significant reduction in NT-proBNP
- Reduction in cardiac troponin I
- Evidence of biological activity

- Reversible reductions in left ventricular ejection fraction (LVEF) in several participants

Although encouraging, these studies did not establish sufficient evidence for approval in non-obstructive HCM.

3.4.3 Phase III: EXPLORER-HCM

The pivotal EXPLORER-HCM trial established the clinical efficacy of mavacamten.

Compared with placebo, patients receiving mavacamten experienced:

- Significant reduction in LVOT gradient
- Improved exercise capacity
- Better NYHA functional class
- Improved Kansas City Cardiomyopathy Questionnaire (KCCQ) scores
- Improved quality of life
- Reduced symptom burden

Subsequent studies involving Chinese patients produced similar findings, confirming reproducibility across populations.

3.5 Therapeutic Applications

Following FDA approval in 2022, mavacamten became indicated for adults with symptomatic NYHA Class II–III obstructive hypertrophic cardiomyopathy.

Unlike conventional medications, mavacamten modifies disease physiology by reducing excessive sarcomeric contractility. Clinical use has demonstrated improvements in exercise tolerance, ventricular outflow obstruction, and patient-reported quality of life.

3.6 Safety Profile

Because mavacamten reduces myocardial contractility, excessive dosing may produce systolic dysfunction.

The principal safety considerations include:

- Reduced left ventricular ejection fraction
- Risk of heart failure
- Need for routine echocardiographic monitoring
- Drug interactions involving CYP2C19 metabolism

The FDA therefore requires treatment through the Risk Evaluation and Mitigation Strategy (REMS) program to ensure appropriate patient selection and dose adjustment.

3.7 Market Impact

Mavacamten represents the first disease-specific pharmacological therapy developed for hypertrophic cardiomyopathy in more than three decades.

Its introduction fundamentally changed therapeutic management by shifting treatment from symptomatic control toward molecular intervention.

Despite the need for specialist monitoring, mavacamten established an entirely new class of cardiac myosin inhibitors and has become the reference therapy against which future agents are compared.

4 Conclusion

The literature consistently demonstrates that mavacamten represents a major advancement in the treatment of obstructive hypertrophic cardiomyopathy. By directly targeting cardiac myosin, the drug addresses the molecular mechanism responsible for hypercontractility rather than merely controlling symptoms. Clinical trials have shown substantial improvements in hemodynamics, functional capacity, and quality of life while maintaining an acceptable safety profile when appropriate monitoring is performed. Future studies will determine its long-term effectiveness, role in non-obstructive disease, and position among emerging myosin inhibitors.

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